

Machine learning

Definition:

"Field of study that gives computers the ability to learn w/o being explicitly programmed"

Algorithms (ML):

- Supervised Learning ← rapid advancements used most in real world apps
- Unsupervised learning
- Recommender systems
- Reinforcement learning

Supervised learning:

99% of economic value by ML is through this.

- refers to algos that learn x to y mappings.

$x \rightarrow y$

input

output label

learns from being given right answers:

↳ correct label y for given input x (correct pairs of x & y).

e.g.: x

y

App

email

spam(0/1)

spam filtering

audio

text trans.
cripts

speech recognition

English

Spanish

machine translation

ad, user info

click?(0/1)

online advertising

img, radar info

pos of other cars

self driving car

img of phone

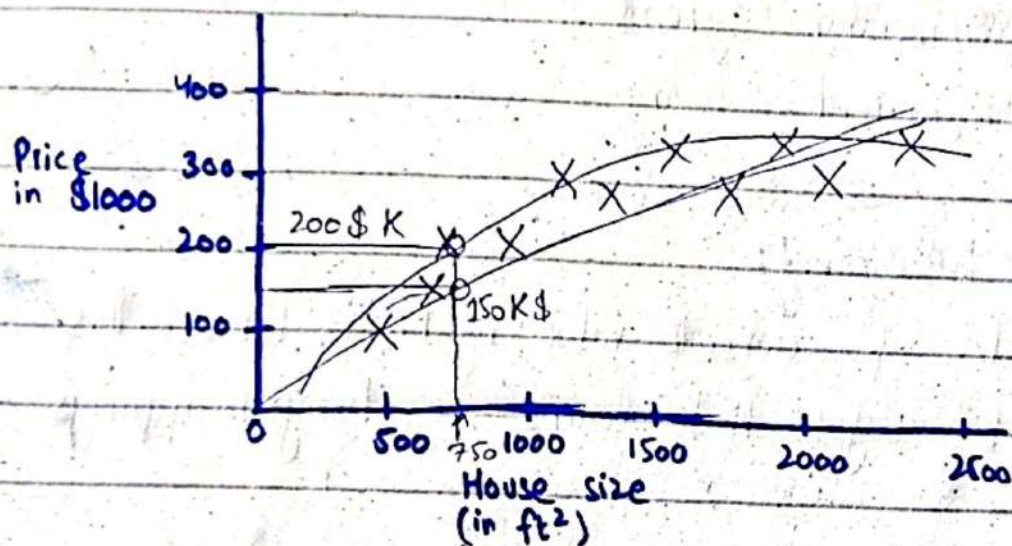
defect(0/1)

visual inspection

- In these, you will first train your model with examples of input (x) & their right answers (y) (x, y pair)
- After learning it will take the brand new input x it has never seen before & try to produce appropriate corresponding output y .

e.g.:

Housing price prediction - Regression



- 750 ft² house price? straight line (150K\$)
curve (200K\$)
- how to decide whether to use straight & curve or something more complex.

Regression:

• Predicting a number from infinitely many possible outputs
like house price could be 150K, 80K, 70K etc.

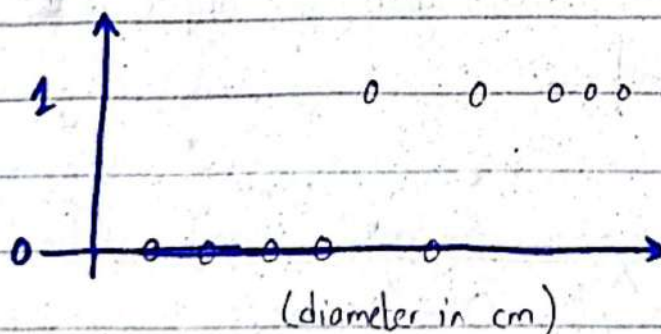
Major type of Supervised learning:

- 1). Regression (what we saw in house prediction)
- 2). Supervised Classification (next).

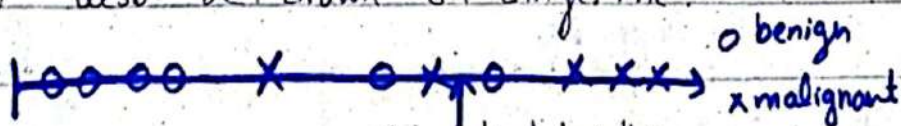
Classification: Breast cancer detection

After seeing patient med records to see whether the lump is malignant or benign.

Size (cm)	diagnosis	
2	0	0 → benign
5	1	1 → malignant
1	0	
7	1	
...		



- The fact there's only two outputs makes it classification.
- can also be drawn on single line.



if a patient has this size tumor? what will it be classified as?

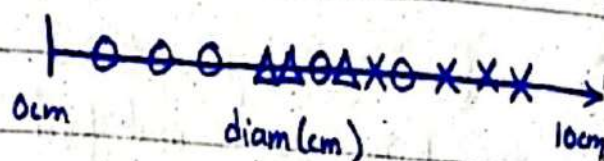
- There can be more than two possible outcomes.
e.g. your machine learning algo can output multiple types of cancers. if it turns out malignant.

Now:

O benign

x malignant type 1

Δ malignant type 2

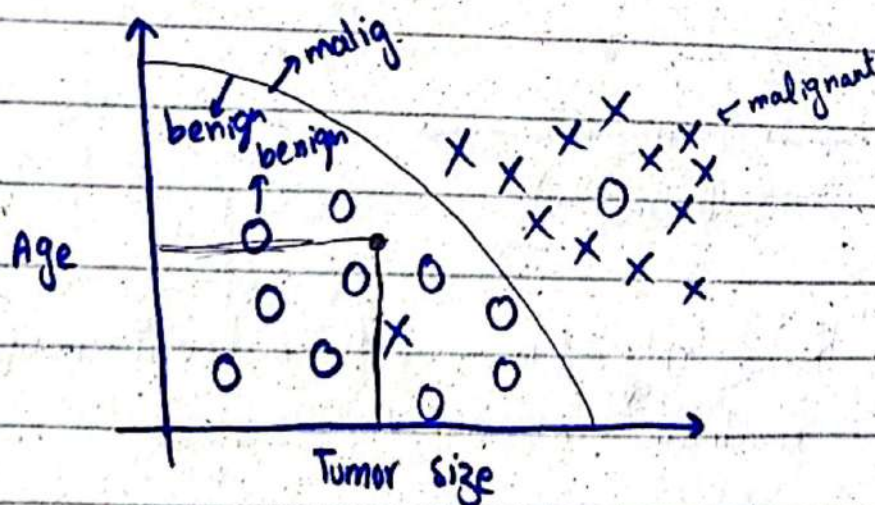


now three possible output class/categories.

- Classification predict categories (can be non-numeric)
i.e. image is cat or dogs, benign or malignant, 0, 1, 2
→ small no. of possible output.

like 0, 1, 2 not 0.5, 1.7 in between ones

We looked at example having only one input value.
But you can also use more than one input value
to predict an output.



what learning algo might do is find some boundary that separates malignant tumor from benign ones.

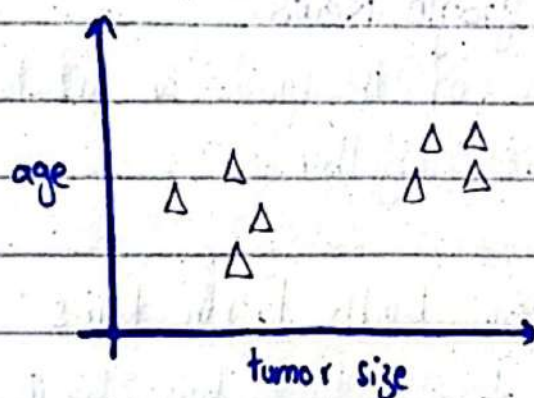
→ So in this the tumor is more likely to be benign

- in many examples more inputs can be used.

e.g. thickness of tumor clump, uniformity of cell size
& so on.

Unsupervised learning:

We are given data that aren't associated with any output labels y .

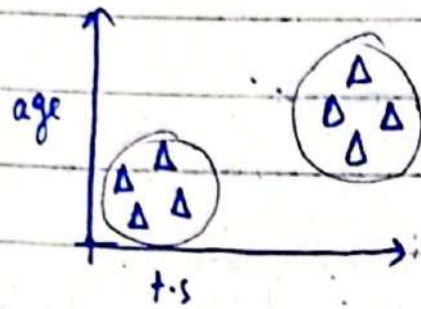


only given data on
patient age or tumor size
but not whether its
benign or malignant.

we are not asked to diagnose whether tumor is benign or not. our job is to find a pattern or just find something interesting in data.

- called unsupervised bcz we are not supervising the algorithm to give some right answer for every input. instead we asked the room to figure out all by yourself what's interesting or what patterns or structure that might be in this dataset.

→ with this particular dataset on unsupervised learning algo might decide the data can be assigned to two different groups or clusters.



Particular type of unsupervised algo called clustering algorithm

e.g.

Clustering: used in google News

it reads hundreds of thousands of articles and group related stories together.

e.g.

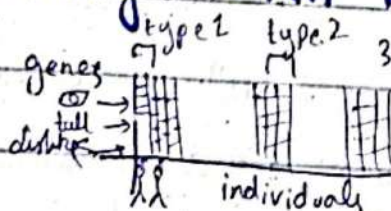
Giant panda give birth to rare twins

• article 1

Panda word & twin also in all articles 300 too.

• 2 etc.

Clustering: DNA micro Array

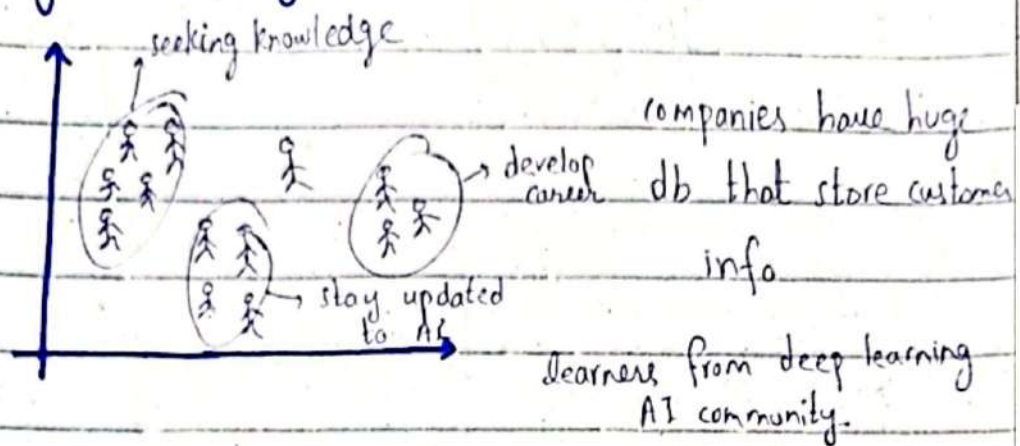


red-green tells whether specific gene active or not.

cluster individuals into diff categories.

→ here's bunch of data I don't know what diff types of people are but can you auto find structure into data.

Clustering: Grouping customers



Group customer auto into diff market segments so that you can more efficiently serve your customers.

Unsupervised formal def:

Data only comes with input x but not output label y . Algo has to find some structure in the data.

Clustering $\rightarrow$ groups together similar data

[Anomaly detection $\rightarrow$ find unusual data points e.g fraud detection
Dimensionality Reduction $\rightarrow$ large dataset compressed using fewer data number with out using much info (as less as possible).]

Dive water